

High-Level Design Document

1. System Overview

The Cryptocurrency Volatility Prediction system is designed to forecast market volatility using historical price data and machine learning techniques. The system helps traders and analysts anticipate market instability and manage risk.

2. System Architecture

The system consists of the following high-level components:

1. Data Source
 2. Data Preprocessing Module
 3. Feature Engineering Module
 4. Machine Learning Model
 5. Prediction Output Module
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3. Input and Output Design

Input:

- Open, High, Low, Close prices
- Trading volume
- Market capitalization

Output:

- Predicted volatility value
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4. Technology Stack

- Programming Language: Python
 - Libraries: Pandas, NumPy, Scikit-learn
 - Visualization: Matplotlib, Seaborn
 - Environment: Jupyter Notebook, VS Code
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5. Advantages of the System

- Modular architecture
 - Scalable and reusable
 - Easy integration with deployment tools
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6. Conclusion

The high-level design provides a clear overview of the system architecture and ensures smooth data flow from input to prediction.